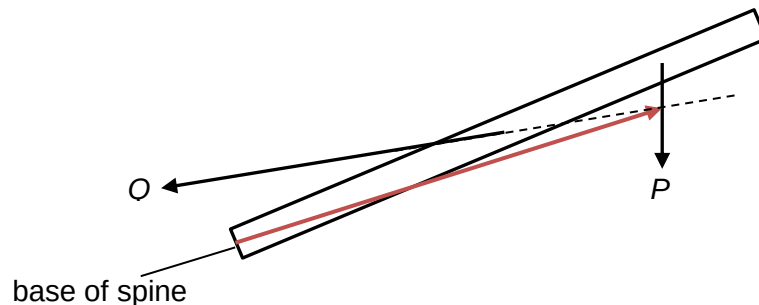


2 (a) (i) The resultant force in any direction is zero. [B1]

The resultant torque about any point/axis is zero. [B1]

(ii)



[B1] arrow from base of spine and correctly going through the point of intersections of all the forces.

(iii) Taking the base of the spine as the pivot,

$$Q \sin 12^\circ (0.40) = P \sin 70^\circ (0.65) \quad [\text{M1}]$$

$$\frac{Q}{P} = \frac{\sin 70^\circ (0.65)}{\sin 12^\circ (0.40)}$$

$$= 7.3$$

[A1]

(b) (i) Vertical direction: $F_{\text{net}} = 0$ as the suitcase did not move in the vertical direction.

$$T_y + N = W$$

$$30 \sin 20^\circ + N = 11(9.81)$$

$$N = 97.6 \text{ N}$$

[M1]

[A1]

(ii) The resultant force acting on the body is proportional to the rate of change of its momentum, and the change takes place in the direction of the resultant force. [B1]

(iii) Horizontal direction: $F_{\text{net}} = ma$ as the suitcase accelerated horizontally.

$$T_x - f = ma$$

$$30 \cos 20^\circ - 8.5 = 11a$$

$$a = 1.79 \text{ m s}^{-2}$$

[M1]

[A1]

3 (a) Rate of change of angular displacement [B1]